Supplementary Material

Rank or threshold? Why fake-news detectors fail on new corpora

Dewan Fahad Chowdhury*
Md. Rejaul Korim Sadi*
Department of Computer Science and Engineering, Metropolitan University, Sylhet, Bangladesh

This supplement documents how the data were prepared and how every number in the paper was produced. All tables here are written by `src/paper_numbers.py` from the result files released with the code; the repository's `run_all.sh` regenerates them end to end.

S1 Data preparation

S1.1 Label harmonisation and verification

Every corpus is mapped to 1 = fake, false, rumour or clickbait. The WELFake release codes 1 = fake although its record documents 0 = fake, 1 = real; our working copy followed the documented legend and is inverted back; LIAR keeps {pants-fire, false} versus {mostly-true, true}; Twitter15/16 keep false versus true. Label direction was verified on the raw files with outlet signatures (`src/label_probe.py`): Table S1.

Table S1: Label-direction probes: number of items matching each source signature under raw label 0 / raw label 1 ("copy" is our working copy of WELFake). A reliable-outlet signature should sit under "real" and a fake-site signature under "fake". ISOT and the WELFake release both code 1 = fake (although the WELFake record's legend says 0 = fake); the WELFake working copy, made by following that legend, has every label inverted.

Signature	Expected	ISOT	WELFake release	WELFake copy
title ends '- The New York Times'	real	0 / 0	6,223 / 0	0 / 6,223
title ends '- Breitbart'	real	0 / 0	2,338 / 2	1 / 2,338
text contains '(Reuters)'	real	20,761 / 7	21,270 / 19	8 / 23
text contains 'Featured image via'	fake	0 / 6,098	0 / 6,261	125 / 0
title contains 'BREAKING'	fake	8 / 614	28 / 1,154	199 / 19
text contains '21st Century Wire'	fake	0 / 629	0 / 1,272	5 / 0
text contains 'Activist Post'	fake	0 / 5	0 / 33	21 / 0

S1.2 Deduplication

Within each corpus, rows are normalised (NFKC, lower case, URLs and punctuation removed), exact duplicates are removed, and near-duplicates are clustered with MinHash-LSH (128 permutations, Jaccard ≥ 0.8; word 3-grams over the first 400 words for articles, character 5-grams otherwise). Any exact or near-duplicate group whose members disagree on the label is removed entirely; otherwise one member is kept. Across corpora, each item is flagged with the corpora in which it has a near-duplicate, and transfer tests drop flagged target items.

*Equal contribution, joint first authors.

Table S2: Deduplication report per corpus.

Corpus	raw	empty	exact conflict	exact dup.	near conflict	near dup.	final
ISOT	38,833	5	0	0	0	79	38,749
WELFake	24,530	1	0	0	6	815	23,708
PolitiFact	1,056	0	4	71	2	3	976
GossipCop	22,140	0	356	1,174	24	26	20,560
LIAR	8,090	0	3	21	0	11	8,055
Twitter15	742	0	0	55	0	14	673
Twitter16	412	0	0	51	0	3	358
PHEME	5,802	0	21	124	4	98	5,555
ClickbaitEN	32,000	0	0	2	0	123	31,875
BanglaBait	15,056	0	23	850	18	124	14,041

S2 Evaluation details

Splits are stratified 70/10/20 per seed (0–4), with caps of 8,000/1,500/3,000 items. Hyperparameters are chosen on the source validation split by ROC-AUC, and scores are Platt-calibrated on the same split. Metrics are computed on the target test split after near-duplicate exclusion. Pooled seed-bootstrap intervals use 200 resamples per seed, pooled across seeds. Operating points: default $t = 0.5$; EM prior adaptation [Saerens et al., 2002] with 200 iterations and tolerance 10^{-6}; rate transfer (threshold at the $(1-\pi_A)$ quantile of the target scores); 0.5-rate (median threshold); a 2-component Gaussian mixture on target logits; the true-prior quantile; adjusted classify-and-count [Forman, 2008] with error rates from the source test split at $t = 0.5$ (falling back to the raw flag rate when $\mathrm{TPR} - \mathrm{FPR} < 0.05$, estimates clipped to $[0.01, 0.99]$); the source in-domain flag rate q_A; and the oracle maximum over 199 quantile thresholds. The cross-fitted oracle tunes the threshold on a random half of the target test set and scores it on the other half, for five random splits and both halves. Corpus-bootstrap intervals resample the seven veracity corpora with replacement (2,000 draws, seed 0) and weight each directed pair of distinct drawn corpora by the product of their multiplicities. The prior-shift experiment resamples each target test set with replacement to 1,000 items at base rate $\pi_A + \delta$, $\delta \in \{-0.3, -0.2, \ldots, 0.3\}$, clipped to $[0.05, 0.95]$, three draws per cell and seed.

Fine-tuned RoBERTa Weights: RoBERTa-base as packaged in spaCy `en_core_web_trf` 3.8.0 (byte-level BPE, 12 layers). The embedding layer and layers 1–8 are frozen; their output for every text is computed once (16-bit floats) and cached. Layers 9–12 and a linear head on the mask-weighted mean of the final states are trained with AdamW (weight decay 0.01, learning rate 3×10^{-5} for the layers and 10^{-3} for the head, linear decay, gradient clipping at 1.0), batch size 16, three epochs, binary cross-entropy with positive-class weight $(1 - \pi_A)/\pi_A$, dropout 0.1. The epoch with the highest validation AUC is kept and its scores are Platt-calibrated on validation. Inputs are the first 1,500 characters, cut to 128 word pieces for articles and 64 for the other units. Each source uses 800 training and 200 validation items and each target 300 test items, stratified subsets of the seed's regular splits, for seeds 0–2. The paired TFIDF-LR sees the first 600 characters of the same items.

S3 Exact values behind the main-text figures

The main text shows these results as figures; the tables below give the exact values.

Table S3: Share of the operating-point loss (oracle minus default macro-F1, off-diagonal cells) recovered by each threshold rule, per detector. Rate rules flag the top share of target items given by the source base rate π_A, the source in-domain flag rate q_A, one half, the adjusted classify-and-count estimate (ACC), or the true target base rate. The last two columns give the mean absolute error of the ACC and EM estimates of the target base rate; the source base rate itself is off by 0.09 on average.

Model	$F1_{0.5}$	$F1^{\star}$	EM	π_A	q_A	0.5	ACC	GMM	π_B	err. ACC	err. EM
TFIDF-LR	0.439	0.565	−64%	81%	76%	80%	−27%	57%	86%	0.41	0.46
TFIDF-SVM	0.437	0.564	−59%	81%	77%	79%	−29%	52%	86%	0.42	0.45
NB-SVM	0.410	0.555	−41%	79%	76%	80%	−16%	24%	85%	0.44	0.48
GLOVE-LR	0.465	0.564	−84%	76%	65%	78%	−56%	29%	83%	0.41	0.43
STYLE-GB	0.427	0.548	−67%	77%	64%	77%	−36%	41%	84%	0.43	0.46
ROBERTA-LR	0.452	0.549	−87%	66%	52%	66%	−49%	25%	75%	0.39	0.43
ROBERTA-FT	0.418	0.571	−51%	77%	67%	76%	−23%	40%	82%	0.42	0.48

Table S4: In-domain versus transfer performance on the seven veracity corpora (mean ± s.d. over seeds of the per-seed mean over cells). "Pairs" averages the eight directed same-task cells. Off-diagonal macro-F1 is given at the default threshold and at the oracle threshold. ROBERTA-LR is evaluated on the 1,500-item subsets and ROBERTA-FT on 800 training and 300 test items per corpus with three seeds (see the paper's Methodology); compare each with its paired TFIDF-LR in the paper, not with the rows above.

		ROC-AUC			macro-F1		
Model	Regime	In-domain	Off-diag.	Pairs	In-domain	Off-diag.	Off-diag. oracle
TFIDF-LR	full	0.914 ±0.006	0.567 ±0.006	0.624 ±0.004	0.854 ±0.005	0.439 ±0.011	0.565 ±0.004
TFIDF-LR	matched	0.861 ±0.010	0.556 ±0.004	0.590 ±0.009	0.786 ±0.012	0.446 ±0.014	0.558 ±0.002
TFIDF-LR	permuted	0.519 ±0.026	0.499 ±0.012	0.514 ±0.019	0.397 ±0.014	0.377 ±0.015	0.518 ±0.007
TFIDF-SVM	full	0.914 ±0.005	0.564 ±0.005	0.625 ±0.005	0.855 ±0.005	0.437 ±0.009	0.564 ±0.004
TFIDF-SVM	matched	0.862 ±0.010	0.554 ±0.005	0.592 ±0.010	0.787 ±0.011	0.447 ±0.017	0.556 ±0.003
TFIDF-SVM	permuted	0.519 ±0.024	0.498 ±0.015	0.511 ±0.020	0.398 ±0.009	0.377 ±0.012	0.516 ±0.008
NB-SVM	full	0.912 ±0.004	0.547 ±0.006	0.590 ±0.011	0.853 ±0.008	0.410 ±0.008	0.555 ±0.003
NB-SVM	matched	0.862 ±0.011	0.540 ±0.006	0.559 ±0.004	0.786 ±0.012	0.398 ±0.013	0.549 ±0.004
NB-SVM	permuted	0.510 ±0.023	0.494 ±0.009	0.502 ±0.009	0.400 ±0.003	0.383 ±0.012	0.520 ±0.006
GLOVE-LR	full	0.864 ±0.005	0.564 ±0.005	0.610 ±0.005	0.782 ±0.007	0.465 ±0.004	0.564 ±0.004
GLOVE-LR	matched	0.830 ±0.006	0.563 ±0.010	0.592 ±0.017	0.740 ±0.011	0.466 ±0.010	0.564 ±0.007
GLOVE-LR	permuted	0.520 ±0.022	0.504 ±0.026	0.494 ±0.021	0.404 ±0.020	0.391 ±0.012	0.525 ±0.019
STYLE-GB	full	0.745 ±0.009	0.535 ±0.004	0.595 ±0.008	0.679 ±0.005	0.427 ±0.007	0.548 ±0.003
STYLE-GB	matched	0.702 ±0.008	0.526 ±0.009	0.561 ±0.032	0.632 ±0.008	0.413 ±0.017	0.542 ±0.007
STYLE-GB	permuted	0.503 ±0.018	0.502 ±0.016	0.493 ±0.028	0.410 ±0.008	0.390 ±0.007	0.523 ±0.011
ROBERTA-LR	full	0.861 ±0.012	0.544 ±0.008	0.630 ±0.011	0.777 ±0.013	0.452 ±0.013	0.549 ±0.010
ROBERTA-FT	full	0.867 ±0.010	0.559 ±0.013	0.614 ±0.020	0.795 ±0.005	0.418 ±0.009	0.571 ±0.008

Table S5: Same-task pairs (TFIDF-LR, five seeds). AUC with pooled seed-bootstrap 95% CI; "In" is the source's in-domain AUC; prior is the target fake rate and PPR the predicted-positive rate at $t = 0.5$; macro-F1 at the default, prior-matched and oracle thresholds; ECE of the source-calibrated scores on the target.

Pair	In	AUC [95% CI]	Prior	PPR	$F1_{0.5}$	$F1_{\pi}$	$F1^{\star}$	ECE
ISOT→WELFake	0.999	0.690 [0.66, 0.71]	0.43	0.78	0.539	0.635	0.638	0.38
WELFake→ISOT	0.989	0.847 [0.82, 0.87]	0.46	0.36	0.753	0.763	0.765	0.11
PolitiFact→GossipCop	0.893	0.526 [0.48, 0.56]	0.23	0.54	0.464	0.514	0.517	0.32
GossipCop→PolitiFact	0.869	0.589 [0.50, 0.68]	0.43	0.07	0.425	0.576	0.595	0.28
Twitter15→Twitter16	0.981	0.672 [0.52, 0.82]	0.46	0.62	0.572	0.597	0.646	0.25
Twitter16→Twitter15	0.967	0.657 [0.55, 0.75]	0.51	0.63	0.580	0.591	0.631	0.28
LIAR→PolitiFact	0.698	0.508 [0.42, 0.60]	0.44	0.54	0.495	0.517	0.536	0.12
PolitiFact→LIAR	0.893	0.505 [0.47, 0.54]	0.44	0.40	0.497	0.502	0.509	0.22

Table S6: Transfer gap decomposition (Eq. 1 of the paper), macro-F1 points, mean over cells and seeds. The gap is relative to the target's own in-domain score.

Model	Cells	Gap	Ranking	Threshold	Prior/cal.	Ranking share
TFIDF-LR	pairs	0.308	0.244	0.018	0.046	79%
TFIDF-LR	other cells	0.441	0.300	0.018	0.122	68%
TFIDF-SVM	pairs	0.308	0.243	0.016	0.049	79%
TFIDF-SVM	other cells	0.444	0.303	0.018	0.123	68%
NB-SVM	pairs	0.319	0.261	0.023	0.036	82%
NB-SVM	other cells	0.472	0.306	0.021	0.145	65%
GLOVE-LR	pairs	0.281	0.193	0.016	0.071	69%
GLOVE-LR	other cells	0.327	0.224	0.017	0.085	69%
STYLE-GB	pairs	0.141	0.091	0.013	0.036	65%
STYLE-GB	other cells	0.278	0.141	0.020	0.117	51%
ROBERTA-LR	pairs	0.243	0.162	0.022	0.059	66%
ROBERTA-LR	other cells	0.344	0.244	0.024	0.076	71%
ROBERTA-FT	pairs	0.340	0.186	0.031	0.123	55%
ROBERTA-FT	other cells	0.387	0.233	0.027	0.127	60%

Table S7: Off-diagonal macro-F1 under different operating points (mean over the 42 cells and five seeds). Label-free: default $t = 0.5$, EM prior adaptation, rate transfer (predicted-positive rate set to the source base rate), predicted-positive rate 0.5, and a 2-component GMM on target logits. With one number known: rate set to the true target prior. Upper bound: oracle threshold. For TFIDF-LR, Holm-corrected Wilcoxon p versus default: EM 1.6e-32, rate transfer 6.6e-31, rate 0.5 1.4e-29, GMM 1.7e-18.

	label-free					prior known	bound
Model	$t = 0.5$	EM	rate transfer	rate 0.5	GMM	rate $= \pi_B$	oracle
TFIDF-LR	0.439	0.358	0.541	0.539	0.510	0.547	0.565
TFIDF-SVM	0.437	0.363	0.539	0.537	0.503	0.546	0.564
NB-SVM	0.410	0.349	0.525	0.526	0.444	0.534	0.555
GLOVE-LR	0.465	0.381	0.540	0.542	0.494	0.547	0.564
STYLE-GB	0.427	0.347	0.521	0.521	0.477	0.529	0.548
ROBERTA-LR	0.452	0.368	0.516	0.516	0.477	0.525	0.549
ROBERTA-FT	0.418	0.340	0.535	0.534	0.479	0.544	0.571

Table S8: Shortcut indicators (ROC-AUC, mean of five seeds). In-domain AUC of TFIDF-LR with the full training split, of the surface-statistics STYLE-GB, and of TFIDF-LR trained on only 64 or 256 items; AUC of a group-identity baseline (publisher domain for PolitiFact and GossipCop, speaker for LIAR); and the mean off-diagonal AUC with the corpus as source and as target.

	in-domain					transfer	
Corpus	full	style only	$n = 64$	$n = 256$	group only	as source	as target
ISOT	0.999	0.999	0.983	0.991	–	0.611	0.670
WELFake	0.989	0.951	0.894	0.941	–	0.626	0.540
PolitiFact	0.893	0.787	0.742	0.850	0.840	0.533	0.592
GossipCop	0.869	0.659	0.610	0.723	0.807	0.538	0.522
LIAR	0.698	0.590	0.567	0.607	0.664	0.537	0.530
Twitter15	0.981	0.603	0.793	0.962	–	0.572	0.563
Twitter16	0.967	0.625	0.890	–	–	0.556	0.552

Table S9: Masking ablation (TFIDF-LR, ROC-AUC, three seeds, first 3,000 characters). Rows: training corpus and what is masked in every corpus; columns: test corpus (the training corpus's own column is in-domain).

Train	Masked	ISOT	WELFake	PolitiFact	GossipCop	LIAR
ISOT	none	0.998	0.679	0.739	0.481	0.559
ISOT	source tags, datelines, URLs, handles	0.997	0.681	0.733	0.483	0.559
ISOT	dates, weekdays, numbers	0.998	0.692	0.723	0.479	0.563
ISOT	named entities	0.996	0.657	0.678	0.543	0.557
ISOT	all of the above	0.996	0.650	0.678	0.540	0.556
WELFake	none	0.808	0.980	0.636	0.529	0.532
WELFake	source tags, datelines, URLs, handles	0.815	0.979	0.634	0.533	0.536
WELFake	dates, weekdays, numbers	0.831	0.977	0.608	0.509	0.526
WELFake	named entities	0.833	0.965	0.614	0.511	0.522
WELFake	all of the above	0.810	0.964	0.611	0.500	0.522

Table S10: Leave-one-corpus-out repairs (TFIDF-LR features; mean over the seven held-out veracity corpora and five seeds; s.d. across seeds). ΔAUC versus pooled training with Holm-corrected Wilcoxon p over the 35 (target, seed) pairs. For reference, the mean single-source transfer AUC to the same targets is 0.567, the best single source (chosen with target labels) 0.671, and the targets' own in-domain AUC 0.914.

Training	AUC	AP	$F1_{0.5}$	F1⋆	ΔAUC (p)
pooled (6 corpora)	0.660 ±0.013	0.582	0.581	0.634	–
+ (corpus, class) balancing	0.659 ±0.013	0.583	0.582	0.632	-0.001 (1.000)
+ balancing + masking	0.661 ±0.012	0.584	0.576	0.633	+0.001 (1.000)
+ balancing + corpus-feature deletion	0.636 ±0.011	0.556	0.560	0.611	-0.023 (0.006)
SVD-300 + balancing	0.647 ±0.009	0.566	0.583	0.624	-0.013 (0.043)
SVD-300 + INLP	0.605 ±0.011	0.520	0.525	0.591	-0.054 (0.001)
SVD-300 + CORAL	0.621 ±0.008	0.539	0.563	0.604	-0.038 (<0.001)

Table S11: Leave-one-corpus-out AUC per held-out corpus (mean of five seeds), against single-source transfer and the in-domain ceiling.

Held out	mean single	best single	pooled	+ masking	INLP	CORAL	in-domain
ISOT	0.670	0.847	0.865	0.890	0.805	0.781	0.999
WELFake	0.540	0.690	0.648	0.657	0.643	0.643	0.989
PolitiFact	0.592	0.730	0.677	0.655	0.600	0.650	0.893
GossipCop	0.522	0.544	0.538	0.546	0.558	0.520	0.869
LIAR	0.530	0.552	0.528	0.528	0.502	0.514	0.698
Twitter15	0.563	0.657	0.647	0.645	0.592	0.626	0.981
Twitter16	0.552	0.678	0.715	0.703	0.536	0.616	0.967

Table S12: Held-out-corpus ROC-AUC of published generalisation methods re-implemented on frozen RoBERTa vectors (leave-one-corpus-out, mean of three seeds).

Method	ISOT	WELFake	PolitiFact	GossipCop	LIAR	Tw15	Tw16	Mean
pooled logistic regression	0.924	0.800	0.705	0.547	0.477	0.602	0.565	0.660
pooled MLP	0.953	0.764	0.588	0.544	0.474	0.574	0.579	0.639
EANN-style adversarial	0.936	0.727	0.569	0.567	0.469	0.586	0.574	0.632
MDFEND-style experts	0.927	0.689	0.501	0.551	0.474	0.595	0.559	0.614
ENDEF-style entity debiasing	0.942	0.760	0.592	0.556	0.474	0.581	0.594	0.643

Table S13: BanglaBait, official split. Clickbait-class F1, precision and recall and accuracy (%), as reported by Mahtab et al. [2023], plus ROC-AUC. Reference values from that paper (as released): LR char 3–5-grams F1 66.28, BanglaBERT 71.72, SS-GAN BanglaBERT 75.13.

Test set	Input	Model	F1	P	R	Acc.	AUC
as released	title	LR, char 3–5	64.37	73.63	57.17	77.72	0.834
as released	title	SVM, char 3–5	63.79	73.37	56.42	77.46	0.836
as released	title	LR, word 1–2	48.51	65.59	38.49	71.25	0.737
as released	title+content	LR, char 3–5	61.29	71.92	53.40	76.26	0.792
as released	title+content	SVM, char 3–5	60.68	71.78	52.55	76.03	0.791
as released	title+content	LR, word 1–2	57.48	70.36	48.58	74.70	0.775
deduplicated	title	LR, char 3–5	59.15	70.72	50.84	76.57	0.823
deduplicated	title	SVM, char 3–5	60.01	71.23	51.85	76.94	0.824
deduplicated	title	LR, word 1–2	40.77	64.56	29.79	71.11	0.722
deduplicated	title+content	LR, char 3–5	52.34	67.98	42.55	74.14	0.772
deduplicated	title+content	SVM, char 3–5	53.28	69.42	43.23	74.70	0.772
deduplicated	title+content	LR, word 1–2	48.61	66.86	38.19	73.06	0.752

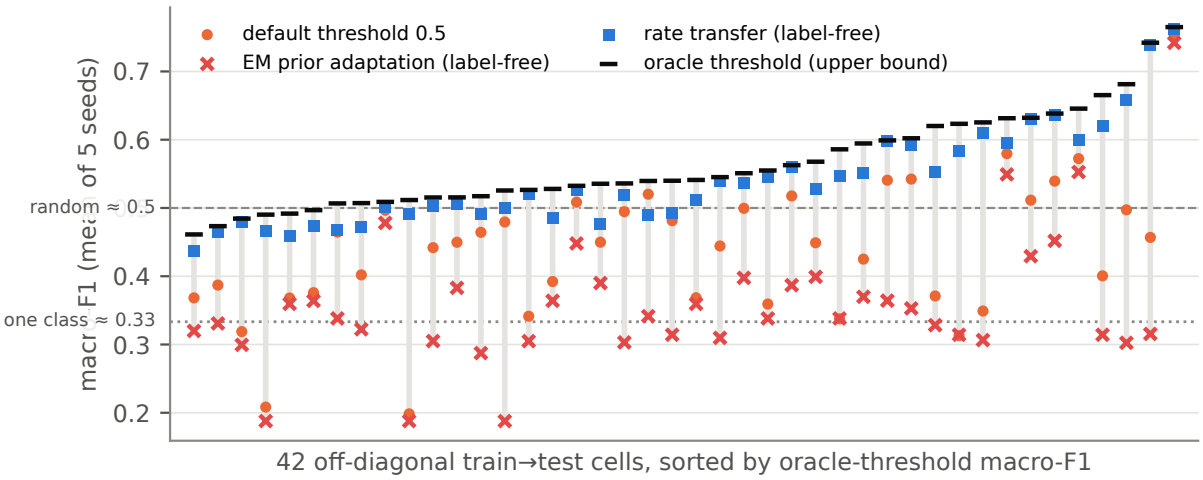


Figure S1: Off-diagonal macro-F1 of TFIDF-LR for each of the 42 cells (mean of five seeds), sorted by the oracle bound. Rate transfer closes most of the distance from the default threshold to the oracle without any target label; EM prior adaptation often falls below the default.

S4 Full results

S4.1 Transfer matrices for every model

Table S14: TFIDF-LR, ROC-AUC (rows: training corpus; columns: test corpus; mean ± s.d. over five seeds).

	ISOT	WELFake	PolitiFact	GossipCop	LIAR	Tw15	Tw16
ISOT	**1.00 ±0.00**	0.69 ±0.01	0.73 ±0.02	0.49 ±0.01	0.55 ±0.02	0.52 ±0.06	0.68 ±0.10
WELFake	0.85 ±0.01	**0.99 ±0.00**	0.66 ±0.02	0.53 ±0.01	0.53 ±0.02	0.58 ±0.03	0.62 ±0.07
PolitiFact	0.80 ±0.01	0.46 ±0.01	**0.89 ±0.02**	0.53 ±0.02	0.50 ±0.01	0.51 ±0.04	0.39 ±0.04
GossipCop	0.48 ±0.04	0.50 ±0.07	0.59 ±0.03	**0.87 ±0.01**	0.54 ±0.02	0.63 ±0.01	0.48 ±0.03
LIAR	0.59 ±0.04	0.63 ±0.02	0.51 ±0.02	0.54 ±0.02	**0.70 ±0.01**	0.48 ±0.05	0.47 ±0.05
Twitter15	0.63 ±0.05	0.51 ±0.03	0.54 ±0.01	0.54 ±0.02	0.54 ±0.02	**0.98 ±0.01**	0.67 ±0.05
Twitter16	0.68 ±0.03	0.46 ±0.01	0.53 ±0.04	0.51 ±0.00	0.51 ±0.02	0.66 ±0.03	**0.97 ±0.02**

Table S15: TFIDF-LR, macro-F1 at $t = 0.5$ (rows: training corpus; columns: test corpus; mean ± s.d. over five seeds).

	ISOT	WELFake	PolitiFact	GossipCop	LIAR	Tw15	Tw16
ISOT	**0.99 ±0.00**	0.54 ±0.01	0.50 ±0.03	0.21 ±0.01	0.44 ±0.03	0.36 ±0.03	0.40 ±0.06
WELFake	0.75 ±0.01	**0.95 ±0.00**	0.35 ±0.03	0.20 ±0.00	0.34 ±0.01	0.34 ±0.00	0.31 ±0.00
PolitiFact	0.46 ±0.03	0.32 ±0.01	**0.81 ±0.02**	0.46 ±0.02	0.50 ±0.01	0.52 ±0.05	0.37 ±0.03
GossipCop	0.37 ±0.02	0.38 ±0.01	0.43 ±0.04	**0.76 ±0.00**	0.37 ±0.01	0.37 ±0.02	0.39 ±0.05
LIAR	0.52 ±0.03	0.54 ±0.03	0.49 ±0.02	0.48 ±0.01	**0.63 ±0.01**	0.46 ±0.03	0.48 ±0.05
Twitter15	0.54 ±0.05	0.44 ±0.03	0.50 ±0.03	0.45 ±0.06	0.51 ±0.03	**0.93 ±0.03**	0.57 ±0.04
Twitter16	0.51 ±0.12	0.39 ±0.07	0.45 ±0.07	0.40 ±0.08	0.45 ±0.06	0.58 ±0.03	**0.92 ±0.03**

Table S16: TFIDF-SVM, ROC-AUC (rows: training corpus; columns: test corpus; mean ± s.d. over five seeds).

	ISOT	WELFake	PolitiFact	GossipCop	LIAR	Tw15	Tw16
ISOT	**1.00 ±0.00**	0.72 ±0.01	0.71 ±0.02	0.50 ±0.02	0.55 ±0.01	0.52 ±0.07	0.68 ±0.10
WELFake	0.85 ±0.01	**0.99 ±0.00**	0.66 ±0.02	0.52 ±0.02	0.53 ±0.02	0.56 ±0.03	0.61 ±0.06
PolitiFact	0.80 ±0.02	0.45 ±0.02	**0.89 ±0.02**	0.52 ±0.02	0.51 ±0.01	0.52 ±0.04	0.39 ±0.05
GossipCop	0.48 ±0.04	0.50 ±0.06	0.59 ±0.03	**0.87 ±0.01**	0.54 ±0.02	0.63 ±0.00	0.48 ±0.04
LIAR	0.58 ±0.04	0.63 ±0.02	0.51 ±0.02	0.55 ±0.02	**0.70 ±0.01**	0.49 ±0.03	0.47 ±0.05
Twitter15	0.58 ±0.06	0.50 ±0.04	0.52 ±0.01	0.53 ±0.02	0.54 ±0.02	**0.98 ±0.01**	0.64 ±0.06
Twitter16	0.68 ±0.03	0.46 ±0.01	0.53 ±0.04	0.51 ±0.01	0.51 ±0.02	0.66 ±0.02	**0.96 ±0.01**

Table S17: TFIDF-SVM, macro-F1 at $t = 0.5$ (rows: training corpus; columns: test corpus; mean ± s.d. over five seeds).

	ISOT	WELFake	PolitiFact	GossipCop	LIAR	Tw15	Tw16
ISOT	**0.99 ±0.00**	0.53 ±0.02	0.42 ±0.03	0.20 ±0.01	0.40 ±0.03	0.35 ±0.01	0.34 ±0.03
WELFake	0.77 ±0.01	**0.96 ±0.00**	0.37 ±0.01	0.21 ±0.00	0.36 ±0.02	0.34 ±0.01	0.31 ±0.00
PolitiFact	0.45 ±0.03	0.32 ±0.01	**0.81 ±0.03**	0.47 ±0.02	0.50 ±0.01	0.53 ±0.04	0.39 ±0.05
GossipCop	0.37 ±0.02	0.37 ±0.01	0.41 ±0.03	**0.76 ±0.00**	0.37 ±0.01	0.37 ±0.02	0.40 ±0.05
LIAR	0.51 ±0.02	0.54 ±0.02	0.50 ±0.02	0.48 ±0.01	**0.64 ±0.01**	0.47 ±0.02	0.49 ±0.04
Twitter15	0.53 ±0.06	0.45 ±0.04	0.49 ±0.03	0.44 ±0.06	0.51 ±0.03	**0.93 ±0.03**	0.56 ±0.03
Twitter16	0.52 ±0.09	0.39 ±0.05	0.46 ±0.06	0.41 ±0.06	0.46 ±0.04	0.59 ±0.03	**0.91 ±0.02**

Table S18: NB-SVM, ROC-AUC (rows: training corpus; columns: test corpus; mean ± s.d. over five seeds).

	ISOT	WELFake	PolitiFact	GossipCop	LIAR	Tw15	Tw16
ISOT	**1.00 ±0.00**	0.54 ±0.01	0.70 ±0.02	0.46 ±0.01	0.54 ±0.02	0.54 ±0.05	0.59 ±0.11
WELFake	0.89 ±0.01	**0.99 ±0.00**	0.64 ±0.02	0.51 ±0.01	0.53 ±0.02	0.55 ±0.02	0.63 ±0.05
PolitiFact	0.75 ±0.02	0.41 ±0.01	**0.90 ±0.01**	0.50 ±0.01	0.50 ±0.01	0.49 ±0.03	0.38 ±0.02
GossipCop	0.38 ±0.02	0.62 ±0.05	0.48 ±0.05	**0.86 ±0.01**	0.55 ±0.02	0.66 ±0.03	0.54 ±0.07
LIAR	0.61 ±0.05	0.54 ±0.02	0.53 ±0.04	0.53 ±0.01	**0.68 ±0.01**	0.40 ±0.07	0.43 ±0.06
Twitter15	0.63 ±0.03	0.40 ±0.03	0.56 ±0.02	0.53 ±0.02	0.52 ±0.01	**0.98 ±0.01**	0.63 ±0.06
Twitter16	0.62 ±0.04	0.42 ±0.04	0.56 ±0.04	0.50 ±0.01	0.51 ±0.01	0.65 ±0.04	**0.97 ±0.01**

Table S19: NB-SVM, macro-F1 at $t = 0.5$ (rows: training corpus; columns: test corpus; mean ± s.d. over five seeds).

	ISOT	WELFake	PolitiFact	GossipCop	LIAR	Tw15	Tw16
ISOT	**1.00 ±0.00**	0.32 ±0.00	0.30 ±0.00	0.19 ±0.00	0.31 ±0.00	0.34 ±0.00	0.31 ±0.00
WELFake	0.81 ±0.02	**0.96 ±0.00**	0.31 ±0.01	0.19 ±0.00	0.31 ±0.00	0.34 ±0.00	0.31 ±0.00
PolitiFact	0.41 ±0.04	0.32 ±0.01	**0.81 ±0.03**	0.47 ±0.01	0.50 ±0.01	0.50 ±0.05	0.39 ±0.03
GossipCop	0.38 ±0.01	0.39 ±0.01	0.43 ±0.04	**0.75 ±0.01**	0.38 ±0.01	0.40 ±0.04	0.42 ±0.05
LIAR	0.57 ±0.04	0.52 ±0.01	0.53 ±0.02	0.51 ±0.01	**0.62 ±0.01**	0.41 ±0.07	0.45 ±0.05
Twitter15	0.45 ±0.05	0.36 ±0.04	0.41 ±0.08	0.33 ±0.10	0.43 ±0.04	**0.92 ±0.03**	0.57 ±0.05
Twitter16	0.50 ±0.05	0.40 ±0.06	0.41 ±0.13	0.34 ±0.10	0.42 ±0.06	0.59 ±0.06	**0.91 ±0.03**

Table S20: GLOVE-LR, ROC-AUC (rows: training corpus; columns: test corpus; mean ± s.d. over five seeds).

	ISOT	WELFake	PolitiFact	GossipCop	LIAR	Tw15	Tw16
ISOT	**0.99 ±0.00**	0.70 ±0.01	0.73 ±0.02	0.48 ±0.01	0.55 ±0.02	0.53 ±0.05	0.69 ±0.03
WELFake	0.81 ±0.01	**0.89 ±0.01**	0.62 ±0.04	0.53 ±0.01	0.53 ±0.02	0.59 ±0.02	0.61 ±0.06
PolitiFact	0.72 ±0.02	0.53 ±0.01	**0.90 ±0.01**	0.56 ±0.01	0.49 ±0.01	0.46 ±0.02	0.36 ±0.03
GossipCop	0.27 ±0.02	0.50 ±0.02	0.55 ±0.03	**0.78 ±0.01**	0.60 ±0.00	0.61 ±0.03	0.50 ±0.08
LIAR	0.61 ±0.06	0.63 ±0.02	0.49 ±0.04	0.65 ±0.01	**0.67 ±0.01**	0.53 ±0.04	0.51 ±0.09
Twitter15	0.54 ±0.03	0.65 ±0.01	0.56 ±0.02	0.49 ±0.02	0.52 ±0.01	**0.91 ±0.01**	0.65 ±0.04
Twitter16	0.67 ±0.04	0.61 ±0.02	0.46 ±0.07	0.51 ±0.02	0.46 ±0.02	0.62 ±0.05	**0.91 ±0.02**

Table S21: GLOVE-LR, macro-F1 at $t = 0.5$ (rows: training corpus; columns: test corpus; mean ± s.d. over five seeds).

	ISOT	WELFake	PolitiFact	GossipCop	LIAR	Tw15	Tw16
ISOT	**0.95 ±0.00**	0.57 ±0.01	0.66 ±0.02	0.23 ±0.00	0.52 ±0.01	0.40 ±0.02	0.51 ±0.05
WELFake	0.68 ±0.01	**0.82 ±0.01**	0.58 ±0.02	0.44 ±0.01	0.53 ±0.02	0.54 ±0.04	0.43 ±0.06
PolitiFact	0.64 ±0.04	0.47 ±0.03	**0.84 ±0.03**	0.30 ±0.04	0.49 ±0.01	0.43 ±0.01	0.34 ±0.05
GossipCop	0.35 ±0.00	0.37 ±0.00	0.41 ±0.03	**0.63 ±0.02**	0.40 ±0.01	0.35 ±0.03	0.37 ±0.03
LIAR	0.56 ±0.05	0.56 ±0.04	0.46 ±0.04	0.36 ±0.03	**0.61 ±0.02**	0.44 ±0.05	0.39 ±0.06
Twitter15	0.40 ±0.02	0.40 ±0.03	0.51 ±0.03	0.44 ±0.02	0.44 ±0.03	**0.83 ±0.03**	0.57 ±0.06
Twitter16	0.59 ±0.03	0.48 ±0.02	0.45 ±0.06	0.49 ±0.01	0.37 ±0.01	0.58 ±0.05	**0.81 ±0.03**

Table S22: STYLE-GB, ROC-AUC (rows: training corpus; columns: test corpus; mean ± s.d. over five seeds).

	ISOT	WELFake	PolitiFact	GossipCop	LIAR	Tw15	Tw16
ISOT	**1.00 ±0.00**	0.79 ±0.02	0.61 ±0.07	0.51 ±0.02	0.55 ±0.02	0.53 ±0.05	0.62 ±0.04
WELFake	0.94 ±0.01	**0.95 ±0.00**	0.48 ±0.05	0.51 ±0.02	0.52 ±0.02	0.58 ±0.04	0.63 ±0.06
PolitiFact	0.64 ±0.04	0.47 ±0.05	**0.79 ±0.01**	0.53 ±0.01	0.49 ±0.01	0.45 ±0.07	0.40 ±0.05
GossipCop	0.40 ±0.05	0.43 ±0.03	0.54 ±0.05	**0.66 ±0.01**	0.55 ±0.02	0.44 ±0.05	0.48 ±0.12
LIAR	0.48 ±0.06	0.61 ±0.07	0.46 ±0.04	0.53 ±0.01	**0.59 ±0.01**	0.55 ±0.03	0.51 ±0.04
Twitter15	0.57 ±0.07	0.60 ±0.06	0.46 ±0.06	0.47 ±0.01	0.50 ±0.02	**0.60 ±0.07**	0.53 ±0.07
Twitter16	0.65 ±0.11	0.51 ±0.04	0.47 ±0.05	0.48 ±0.03	0.50 ±0.03	0.49 ±0.05	**0.63 ±0.07**

Table S23: STYLE-GB, macro-F1 at $t = 0.5$ (rows: training corpus; columns: test corpus; mean ± s.d. over five seeds).

	ISOT	WELFake	PolitiFact	GossipCop	LIAR	Tw15	Tw16
ISOT	**0.99 ±0.00**	0.72 ±0.02	0.34 ±0.07	0.20 ±0.03	0.35 ±0.04	0.35 ±0.01	0.33 ±0.02
WELFake	0.84 ±0.03	**0.87 ±0.01**	0.32 ±0.01	0.21 ±0.01	0.47 ±0.03	0.39 ±0.05	0.38 ±0.04
PolitiFact	0.47 ±0.03	0.43 ±0.01	**0.72 ±0.02**	0.48 ±0.01	0.47 ±0.01	0.47 ±0.06	0.42 ±0.07
GossipCop	0.39 ±0.06	0.41 ±0.05	0.41 ±0.01	**0.54 ±0.01**	0.40 ±0.02	0.35 ±0.03	0.37 ±0.03
LIAR	0.39 ±0.01	0.40 ±0.01	0.44 ±0.04	0.38 ±0.01	**0.49 ±0.03**	0.44 ±0.03	0.44 ±0.06
Twitter15	0.49 ±0.08	0.51 ±0.05	0.39 ±0.03	0.34 ±0.05	0.48 ±0.01	**0.56 ±0.04**	0.53 ±0.07
Twitter16	0.52 ±0.10	0.48 ±0.03	0.44 ±0.05	0.35 ±0.06	0.50 ±0.02	0.46 ±0.06	**0.59 ±0.05**

Table S24: ROBERTA-LR, ROC-AUC (rows: training corpus; columns: test corpus; mean ± s.d. over five seeds).

	ISOT	WELFake	PolitiFact	GossipCop	LIAR	Tw15	Tw16
ISOT	**1.00 ±0.00**	0.72 ±0.04	0.55 ±0.01	0.49 ±0.01	0.52 ±0.05	0.52 ±0.02	0.53 ±0.05
WELFake	0.93 ±0.01	**0.96 ±0.00**	0.47 ±0.04	0.47 ±0.05	0.46 ±0.03	0.62 ±0.03	0.58 ±0.07
PolitiFact	0.78 ±0.04	0.60 ±0.03	**0.90 ±0.02**	0.55 ±0.01	0.49 ±0.03	0.47 ±0.03	0.48 ±0.08
GossipCop	0.64 ±0.08	0.41 ±0.05	0.61 ±0.03	**0.77 ±0.04**	0.55 ±0.03	0.45 ±0.05	0.51 ±0.06
LIAR	0.66 ±0.10	0.51 ±0.09	0.61 ±0.08	0.51 ±0.03	**0.61 ±0.03**	0.49 ±0.02	0.54 ±0.04
Twitter15	0.39 ±0.06	0.56 ±0.05	0.48 ±0.06	0.53 ±0.07	0.49 ±0.04	**0.85 ±0.04**	0.57 ±0.07
Twitter16	0.61 ±0.08	0.57 ±0.03	0.42 ±0.03	0.50 ±0.03	0.45 ±0.06	0.55 ±0.05	**0.93 ±0.01**

Table S25: ROBERTA-LR, macro-F1 at $t = 0.5$ (rows: training corpus; columns: test corpus; mean ± s.d. over five seeds).

	ISOT	WELFake	PolitiFact	GossipCop	LIAR	Tw15	Tw16
ISOT	**0.99 ±0.01**	0.56 ±0.03	0.37 ±0.01	0.20 ±0.01	0.51 ±0.03	0.42 ±0.01	0.44 ±0.06
WELFake	0.79 ±0.04	**0.89 ±0.01**	0.41 ±0.04	0.28 ±0.07	0.46 ±0.03	0.41 ±0.04	0.35 ±0.02
PolitiFact	0.50 ±0.07	0.42 ±0.05	**0.83 ±0.03**	0.42 ±0.03	0.48 ±0.04	0.42 ±0.03	0.36 ±0.03
GossipCop	0.49 ±0.09	0.42 ±0.02	0.46 ±0.03	**0.58 ±0.05**	0.46 ±0.06	0.36 ±0.03	0.38 ±0.04
LIAR	0.55 ±0.11	0.51 ±0.04	0.56 ±0.10	0.40 ±0.03	**0.55 ±0.03**	0.46 ±0.01	0.54 ±0.04
Twitter15	0.42 ±0.05	0.53 ±0.03	0.46 ±0.04	0.49 ±0.05	0.42 ±0.05	**0.75 ±0.02**	0.54 ±0.07
Twitter16	0.52 ±0.13	0.49 ±0.10	0.44 ±0.03	0.38 ±0.07	0.40 ±0.02	0.52 ±0.05	**0.85 ±0.03**

Table S26: ROBERTA-FT, ROC-AUC (rows: training corpus; columns: test corpus; mean ± s.d. over five seeds).

	ISOT	WELFake	PolitiFact	GossipCop	LIAR	Tw15	Tw16
ISOT	**1.00 ±0.00**	0.74 ±0.03	0.69 ±0.03	0.51 ±0.03	0.54 ±0.02	0.55 ±0.03	0.62 ±0.07
WELFake	0.94 ±0.02	**1.00 ±0.00**	0.40 ±0.08	0.52 ±0.06	0.46 ±0.05	0.58 ±0.03	0.55 ±0.06
PolitiFact	0.75 ±0.04	0.47 ±0.01	**0.93 ±0.01**	0.54 ±0.02	0.49 ±0.03	0.43 ±0.03	0.37 ±0.06
GossipCop	0.57 ±0.17	0.47 ±0.05	0.70 ±0.08	**0.78 ±0.05**	0.57 ±0.07	0.48 ±0.04	0.62 ±0.13
LIAR	0.52 ±0.10	0.53 ±0.03	0.37 ±0.04	0.58 ±0.02	**0.71 ±0.02**	0.53 ±0.04	0.53 ±0.05
Twitter15	0.63 ±0.07	0.63 ±0.02	0.54 ±0.09	0.52 ±0.04	0.52 ±0.01	**0.81 ±0.02**	0.57 ±0.00
Twitter16	0.75 ±0.10	0.58 ±0.06	0.41 ±0.07	0.58 ±0.02	0.57 ±0.05	0.55 ±0.05	**0.85 ±0.05**

Table S27: ROBERTA-FT, macro-F1 at $t = 0.5$ (rows: training corpus; columns: test corpus; mean ± s.d. over five seeds).

	ISOT	WELFake	PolitiFact	GossipCop	LIAR	Tw15	Tw16
ISOT	**1.00 ±0.00**	0.51 ±0.03	0.38 ±0.03	0.19 ±0.01	0.37 ±0.05	0.33 ±0.01	0.35 ±0.01
WELFake	0.61 ±0.11	**0.96 ±0.02**	0.38 ±0.04	0.38 ±0.08	0.47 ±0.02	0.37 ±0.03	0.36 ±0.03
PolitiFact	0.34 ±0.00	0.32 ±0.04	**0.85 ±0.01**	0.40 ±0.04	0.37 ±0.03	0.39 ±0.02	0.35 ±0.04
GossipCop	0.36 ±0.01	0.38 ±0.01	0.44 ±0.05	**0.64 ±0.01**	0.41 ±0.06	0.34 ±0.02	0.39 ±0.04
LIAR	0.49 ±0.07	0.53 ±0.02	0.35 ±0.02	0.28 ±0.06	**0.62 ±0.01**	0.43 ±0.06	0.39 ±0.10
Twitter15	0.49 ±0.05	0.55 ±0.05	0.52 ±0.07	0.52 ±0.04	0.48 ±0.02	**0.73 ±0.02**	0.51 ±0.01
Twitter16	0.57 ±0.06	0.49 ±0.13	0.41 ±0.03	0.36 ±0.10	0.50 ±0.12	0.50 ±0.06	**0.77 ±0.03**

S4.2 Per-cell decomposition

Table S28: Per-cell transfer gap decomposition, TFIDF-LR, mean of five seeds (macro-F1 points). Gap = target in-domain F1 minus transfer F1 at $t = 0.5$ = ranking + threshold + prior; "difficulty" is source minus target in-domain F1; "EM" is the change produced by EM prior adaptation.

Source	Target	Gap	Ranking	Threshold	Prior	Difficulty	EM
GossipCop	ISOT	0.620	0.497	0.003	0.121	-0.230	-0.009
GossipCop	LIAR	0.263	0.091	0.006	0.166	0.127	-0.009
GossipCop	PolitiFact	0.382	0.213	0.018	0.151	-0.049	-0.055
GossipCop	Twitter15	0.555	0.306	0.038	0.211	-0.168	-0.043
GossipCop	Twitter16	0.527	0.391	0.076	0.059	-0.161	-0.028
GossipCop	WELFake	0.575	0.454	0.005	0.116	-0.193	-0.012
ISOT	GossipCop	0.550	0.268	0.005	0.277	0.230	-0.020
ISOT	LIAR	0.187	0.086	0.005	0.095	0.357	-0.135
ISOT	PolitiFact	0.310	0.126	0.026	0.158	0.181	-0.195
ISOT	Twitter15	0.567	0.371	0.029	0.166	0.062	-0.021
ISOT	Twitter16	0.518	0.254	0.046	0.219	0.069	-0.086
ISOT	WELFake	0.411	0.312	0.003	0.096	0.038	-0.087
LIAR	GossipCop	0.279	0.233	0.005	0.041	-0.127	-0.292
LIAR	ISOT	0.471	0.426	0.003	0.042	-0.357	-0.131
LIAR	PolitiFact	0.313	0.271	0.019	0.022	-0.176	-0.191
LIAR	Twitter15	0.461	0.419	0.040	0.002	-0.294	-0.127
LIAR	Twitter16	0.438	0.379	0.055	0.004	-0.287	-0.167
LIAR	WELFake	0.410	0.352	0.003	0.055	-0.319	-0.176
PolitiFact	GossipCop	0.294	0.241	0.004	0.049	0.049	-0.177
PolitiFact	ISOT	0.531	0.246	0.003	0.282	-0.181	-0.141
PolitiFact	LIAR	0.135	0.123	0.007	0.005	0.176	-0.018
PolitiFact	Twitter15	0.406	0.386	0.046	-0.027	-0.119	-0.179
PolitiFact	Twitter16	0.551	0.458	0.026	0.067	-0.112	-0.048
PolitiFact	WELFake	0.632	0.466	0.005	0.161	-0.143	-0.019
Twitter15	GossipCop	0.308	0.223	0.003	0.083	0.168	-0.060
Twitter15	ISOT	0.446	0.386	0.005	0.054	-0.062	-0.189
Twitter15	LIAR	0.123	0.099	0.006	0.018	0.294	-0.060
Twitter15	PolitiFact	0.308	0.256	0.027	0.025	0.119	-0.102
Twitter15	Twitter16	0.347	0.273	0.048	0.025	0.007	-0.020
Twitter15	WELFake	0.509	0.435	0.003	0.071	-0.025	-0.137
Twitter16	GossipCop	0.356	0.251	0.004	0.101	0.161	-0.080
Twitter16	ISOT	0.477	0.356	0.002	0.118	-0.069	-0.082
Twitter16	LIAR	0.182	0.116	0.009	0.056	0.287	-0.067
Twitter16	PolitiFact	0.358	0.240	0.031	0.087	0.112	-0.050
Twitter16	Twitter15	0.346	0.294	0.041	0.011	-0.007	-0.030
Twitter16	WELFake	0.564	0.477	0.006	0.081	-0.032	-0.056
WELFake	GossipCop	0.560	0.246	0.003	0.311	0.193	-0.011
WELFake	ISOT	0.235	0.223	0.002	0.010	-0.038	-0.011
WELFake	LIAR	0.290	0.105	0.005	0.180	0.319	-0.036

Source	Target	Gap	Ranking	Threshold	Prior	Difficulty	EM
WELFake	PolitiFact	0.458	0.182	0.014	0.262	0.143	-0.043
WELFake	Twitter15	0.588	0.340	0.028	0.221	0.025	+0.001
WELFake	Twitter16	0.606	0.296	0.043	0.267	0.032	+0.001

S4.3 Per-cell operating points

Table S29: Per-cell macro-F1 under each operating point, TFIDF-LR, mean of five seeds.

Source	Target	$t = 0.5$	EM	rank	0.5 rate	GMM	π_B	oracle
GossipCop	ISOT	0.368	0.360	0.460	0.487	0.487	0.489	0.492
GossipCop	LIAR	0.368	0.359	0.512	0.530	0.520	0.535	0.541
GossipCop	PolitiFact	0.425	0.370	0.552	0.573	0.529	0.576	0.595
GossipCop	Twitter15	0.371	0.328	0.552	0.578	0.579	0.582	0.620
GossipCop	Twitter16	0.392	0.364	0.485	0.457	0.465	0.452	0.528
GossipCop	WELFake	0.376	0.364	0.474	0.493	0.493	0.492	0.497
ISOT	GossipCop	0.208	0.188	0.466	0.456	0.422	0.485	0.490
ISOT	LIAR	0.444	0.310	0.539	0.537	0.540	0.540	0.545
ISOT	PolitiFact	0.497	0.302	0.658	0.667	0.562	0.655	0.681
ISOT	Twitter15	0.359	0.338	0.545	0.530	0.499	0.526	0.555
ISOT	Twitter16	0.401	0.314	0.620	0.624	0.593	0.620	0.665
ISOT	WELFake	0.539	0.452	0.636	0.633	0.629	0.635	0.638
LIAR	GossipCop	0.479	0.188	0.501	0.487	0.402	0.521	0.526
LIAR	ISOT	0.518	0.387	0.559	0.560	0.553	0.560	0.563
LIAR	PolitiFact	0.495	0.303	0.519	0.502	0.475	0.517	0.536
LIAR	Twitter15	0.465	0.338	0.468	0.465	0.443	0.466	0.507
LIAR	Twitter16	0.481	0.314	0.492	0.491	0.375	0.485	0.540
LIAR	WELFake	0.541	0.364	0.597	0.592	0.577	0.596	0.599
PolitiFact	GossipCop	0.464	0.287	0.491	0.474	0.480	0.514	0.517
PolitiFact	ISOT	0.457	0.316	0.739	0.733	0.714	0.739	0.742
PolitiFact	LIAR	0.497	0.478	0.501	0.501	0.498	0.502	0.509
PolitiFact	Twitter15	0.520	0.342	0.489	0.486	0.492	0.493	0.540
PolitiFact	Twitter16	0.368	0.320	0.438	0.418	0.432	0.435	0.461
PolitiFact	WELFake	0.319	0.300	0.479	0.469	0.415	0.480	0.484
Twitter15	GossipCop	0.450	0.390	0.476	0.479	0.513	0.533	0.535
Twitter15	ISOT	0.542	0.353	0.592	0.594	0.588	0.597	0.602
Twitter15	LIAR	0.509	0.448	0.527	0.528	0.525	0.526	0.532
Twitter15	PolitiFact	0.499	0.398	0.536	0.538	0.501	0.524	0.551
Twitter15	Twitter16	0.572	0.553	0.600	0.602	0.517	0.597	0.646
Twitter15	WELFake	0.442	0.305	0.503	0.505	0.493	0.512	0.515
Twitter16	GossipCop	0.402	0.322	0.472	0.462	0.426	0.503	0.507
Twitter16	ISOT	0.511	0.429	0.630	0.628	0.603	0.630	0.632
Twitter16	LIAR	0.450	0.383	0.506	0.503	0.506	0.506	0.516
Twitter16	PolitiFact	0.449	0.399	0.528	0.527	0.511	0.536	0.568
Twitter16	Twitter15	0.580	0.549	0.595	0.590	0.562	0.591	0.631
Twitter16	WELFake	0.387	0.331	0.465	0.463	0.414	0.468	0.473

Source	Target	$t = 0.5$	EM	rank	0.5 rate	GMM	π_B	oracle
WELFake	GossipCop	0.198	0.188	0.491	0.474	0.366	0.509	0.512
WELFake	ISOT	0.753	0.742	0.762	0.759	0.757	0.763	0.765
WELFake	LIAR	0.341	0.305	0.521	0.520	0.490	0.521	0.526
WELFake	PolitiFact	0.349	0.306	0.610	0.603	0.407	0.612	0.625
WELFake	Twitter15	0.338	0.338	0.547	0.557	0.543	0.558	0.586
WELFake	Twitter16	0.313	0.314	0.584	0.574	0.535	0.580	0.623

S4.4 Learning curves

Table S30: Learning curves: in-domain ROC-AUC of TFIDF-LR trained on n items (mean of five seeds).

Corpus	32	64	128	256	512	1024	2048	4096
ISOT	0.973	0.983	0.988	0.991	0.994	0.996	0.998	0.999
WELFake	0.850	0.894	0.927	0.941	0.956	0.968	0.977	0.984
PolitiFact	0.653	0.742	0.799	0.850	0.885	–	–	–
GossipCop	0.563	0.610	0.651	0.723	0.767	0.803	0.832	0.850
LIAR	0.568	0.567	0.585	0.607	0.631	0.654	0.671	0.691
Twitter15	0.670	0.793	0.897	0.962	–	–	–	–
Twitter16	0.810	0.890	0.949	–	–	–	–	–
PHEME	0.657	0.734	0.784	0.838	0.858	0.888	0.912	–
ClickbaitEN	0.823	0.864	0.916	0.945	0.967	0.979	0.987	0.991

S4.5 Vocabulary ablation

Table S31: Vocabulary and n-gram ablation (TFIDF-LR, ROC-AUC, three seeds).

Train	Features	In-domain	Pair	All targets
ISOT	word(1, 1)-1000	0.999	0.703	0.604
ISOT	word(1, 1)-5000	0.999	0.677	0.613
ISOT	word(1, 1)-20000	0.998	0.679	0.614
ISOT	word(1, 2)-100000	0.998	0.675	0.614
ISOT	word(1, 3)-200000	0.998	0.663	0.609
ISOT	char(3, 5)-100000	1.000	0.734	0.620
WELFake	word(1, 1)-1000	0.975	0.783	0.608
WELFake	word(1, 1)-5000	0.980	0.811	0.615
WELFake	word(1, 1)-20000	0.982	0.813	0.624
WELFake	word(1, 2)-100000	0.981	0.823	0.627
WELFake	word(1, 3)-200000	0.981	0.819	0.624
WELFake	char(3, 5)-100000	0.988	0.811	0.608

Table S32: BanglaBait leave-one-publisher-out (character 3–5-gram LR on titles): each row is a held-out publisher.

Publisher	n	clickbait rate	accuracy	AUC	predicted rate
twentyfourbd.com	2,584	0.37	0.674	0.709	0.35
topdhaka.com	1,732	0.46	0.735	0.819	0.29
rtvonline.com	1,071	0.07	0.920	0.882	0.08
nbtimes24.com	967	0.23	0.775	0.715	0.08
kureghornews.com	950	0.36	0.786	0.833	0.20
newzcitizen.com	907	0.34	0.750	0.799	0.30
authoritynewz.com	842	0.40	0.678	0.706	0.35
citynewszet.com	841	0.46	0.772	0.854	0.48
thebasenewz.com	724	0.66	0.715	0.841	0.46
newzauthority.com	531	0.47	0.627	0.704	0.29
newsholder21.com	453	0.25	0.792	0.751	0.16
somoynews.tv	414	0.09	0.908	0.813	0.06
techzoom.tv	375	0.13	0.864	0.787	0.01
kalerkantho.com	321	0.11	0.879	0.797	0.10
channeldhaka.news	321	0.12	0.869	0.783	0.15
earki.com	280	0.30	0.782	0.793	0.26
motikontho.wordpress.com	195	0.08	0.918	0.651	0.01
bengalbeats.com	192	0.19	0.891	0.911	0.18
other (<100 items each)	180	0.51	0.789	0.889	0.37
beanibazarview24.com	161	0.32	0.764	0.787	0.19

S4.6 Bengali: per-publisher results

S5 Additional figures

S6 Code map

`src/prep.py`	corpora, label harmonisation, deduplication, cross-corpus flags
`src/label_probe.py`	label-direction probes
`src/transfer.py`	transfer matrices (5 models × 3 regimes × 5 seeds)
`src/roberta.py`	RoBERTa representations and paired TF-IDF baseline
`src/mechanism.py`	style, group, learning-curve, top-K, masking, topic, PAD, weights, vocabulary
`src/fix.py`	leave-one-corpus-out repairs
`src/bangla.py`	BanglaBait replication, leave-one-publisher-out, translate-test
`src/analysis.py`, `analysis_extra.py`	aggregation, intervals, decomposition, tests
`src/figures.py`, `paper_numbers.py`	every figure, number and table

References

George Forman. Quantifying counts and costs via classification. *Data Mining and Knowledge Discovery*, 17(2): 164–206, 2008. doi: 10.1007/s10618-008-0097-y.

Md. Motahar Mahtab, Monirul Haque, Mehedi Hasan, and Farig Sadeque. BanglaBait: Semi-supervised adversarial approach for clickbait detection on Bangla clickbait dataset. In *Proceedings of the 14th International Conference on Recent Advances in Natural Language Processing (RANLP)*, pages 748–758, 2023. URL `https://aclanthology.org/2023.ranlp-1.81`.

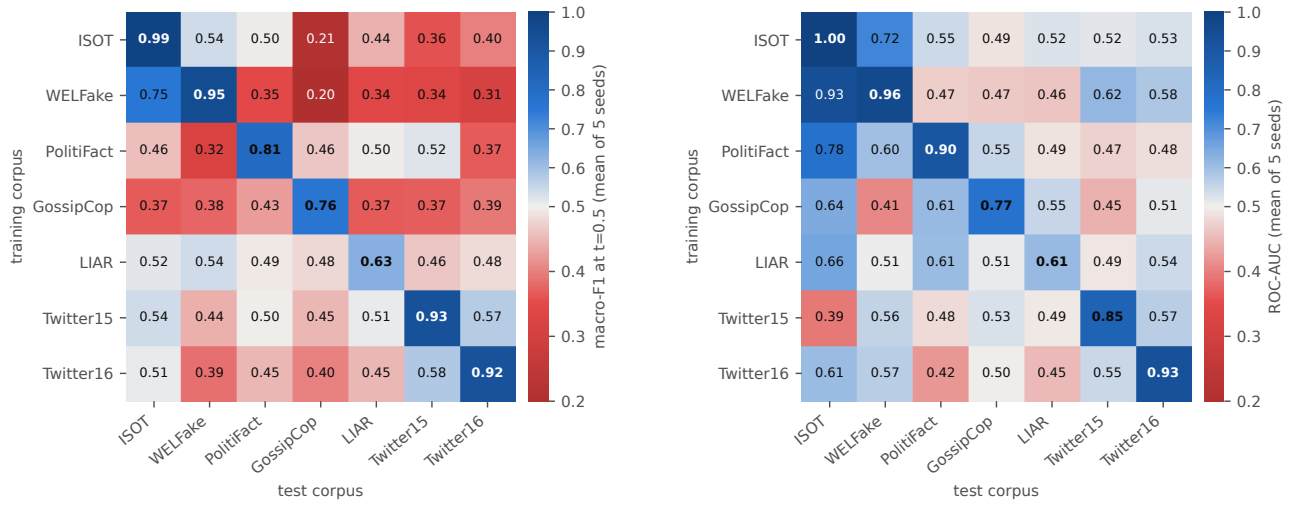


Figure S2: Left: macro-F1 at $t = 0.5$ for TFIDF-LR. Right: ROC-AUC for frozen RoBERTa representations on the 1,500-item subsets.

Marco Saerens, Patrice Latinne, and Christine Decaestecker. Adjusting the outputs of a classifier to new a priori probabilities: A simple procedure. *Neural Computation*, 14(1):21–41, 2002. doi: 10.1162/089976602753284446.

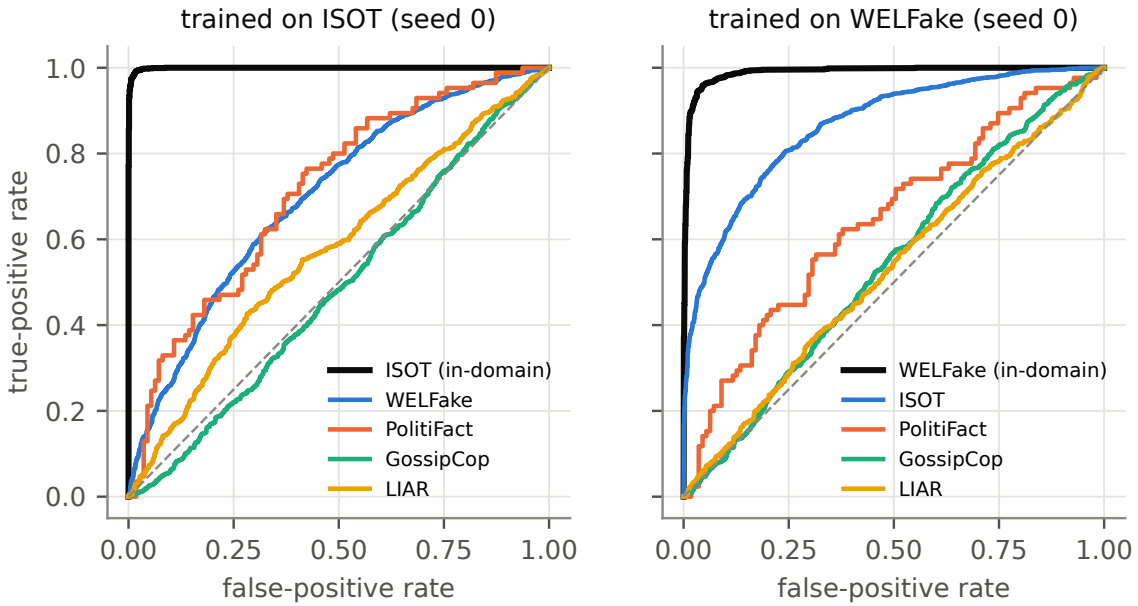


Figure S3: ROC curves (seed 0) for models trained on ISOT and WELFake, in-domain and on four other corpora.

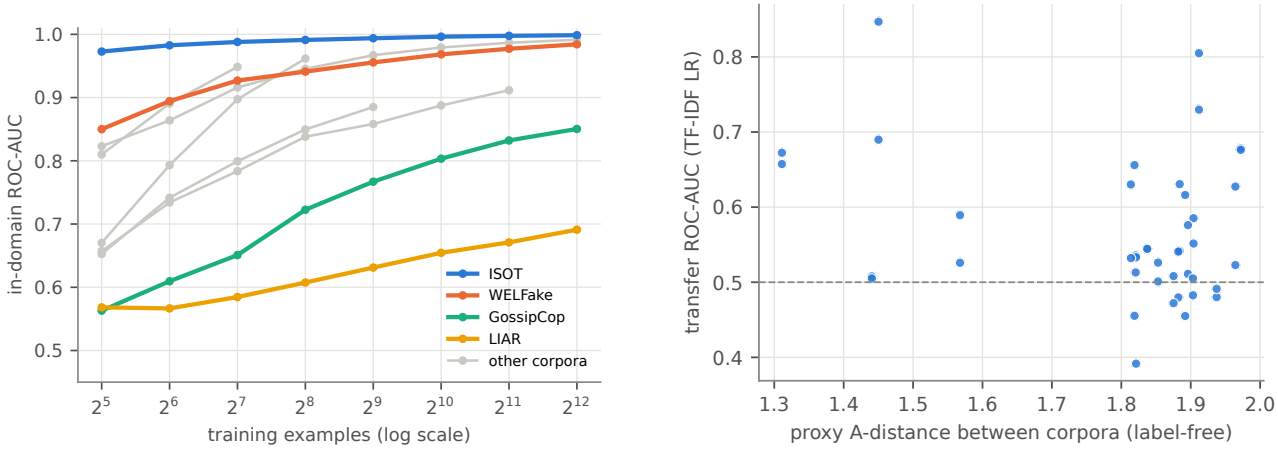


Figure S4: Left: in-domain learning curves. Right: proxy $\mathcal{A}$-distance between corpora against transfer AUC (Spearman $\rho = -0.01$).

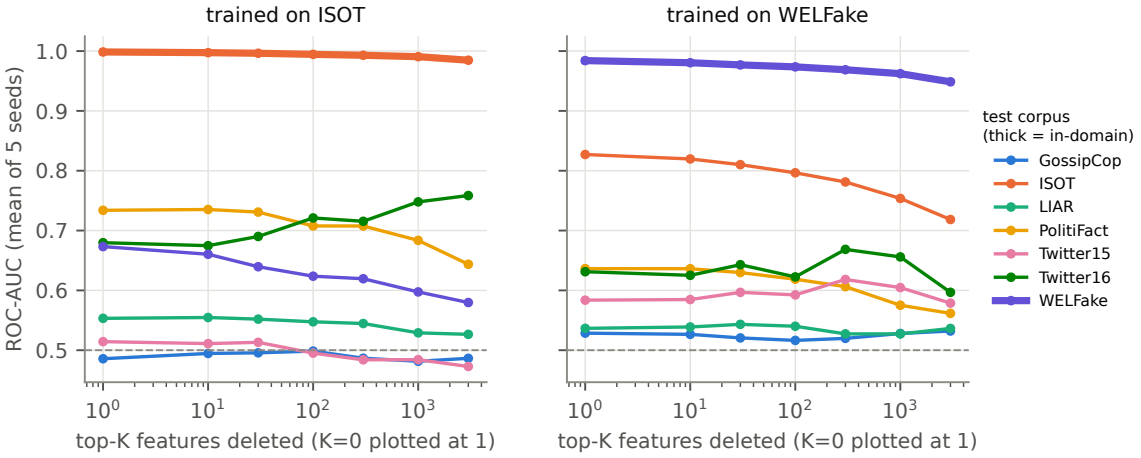


Figure S5: Top-K feature deletion: in-domain and transfer AUC as the K highest-weight features are removed.

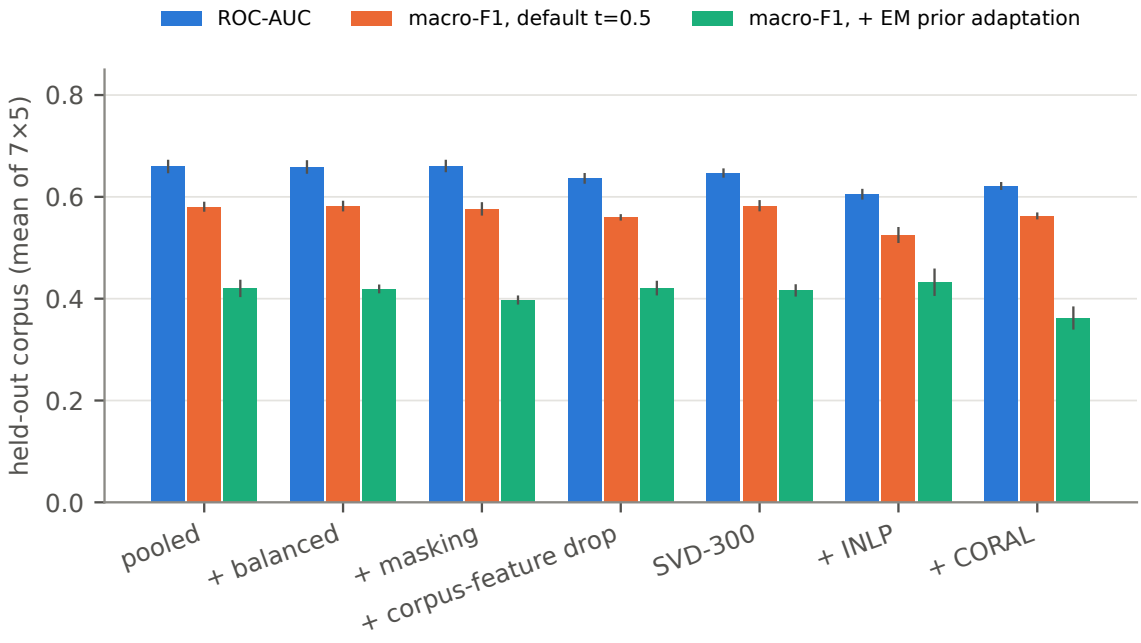


Figure S6: Leave-one-corpus-out repairs: mean over held-out corpora and seeds.